Electrical Instrumentation Part-II
PowerPack Phase 6
Final Semester Reverse-Engineering Master Plan

Last 2-Day Strategy, Question Recognition, Diagram Discipline, Answer Templates
Grassroot-to-exam execution style

Semester Command

This Phase 6 is not a new chapter. This is the final reverse-engineering phase. Its purpose is to convert all learned topics into semester-answer ability. The main target is:

See question → identify topic → draw clean diagram → write definitions → derive → finish with re

Strict Diagram Rule

For every diagram:

1. Use only straight and perpendicular lines.
2. Do not place words on wires.
3. Do not place two labels at the same height if they are close.
4. If one label is horizontal, shift the next label up, down, left or right.
5. Long labels must be split into two or three lines inside a box.
6. If a diagram becomes too large, draw it in two separate figures.

Contents

1. Phase 6 Target

After completing this phase, you should be able to answer by yourself:

1. CRO questions from Lissajous, matched probe, delay line and time-base generator.
2. PLL questions from block diagram, linear model, demodulator and frequency synthesizer.
3. Active filter questions from Butterworth, Chebyshev, state-variable and switched-capacitor filters.
4. DAC and ADC questions from R-2R ladder, converter errors, SAR ADC and dual-slope ADC.
5. Display-device questions from CRT, LED, seven-segment and LCD.
6. Mixed numerical questions using formula-first reverse engineering.
7. OR-type questions by choosing the answer that can be written with better diagram and derivation.

Must Know

In the semester exam, do not try to remember paragraphs. Remember the answer structure. If the structure is clear, the answer can be rebuilt inside the exam hall.

2. The Reverse-Engineering Method

A semester question is not random. It usually asks for one of the following:

definition	diagram	working	derivation	comparison	numerical

Therefore, every answer should be reverse-engineered from the marks.

2.1. Marks-Based Answer Depth

Marks	Required Answer Style
2 marks	Definition plus one key formula or one small explanation.
3 marks	Definition, small diagram if possible, and two to three key points.
4 marks	Definition, neat diagram, working and final formula.
5 marks	Full answer: definition, diagram, working, derivation or explanation and result.
8 marks	Broad answer: introduction, diagram, block-by-block explanation, derivation, physical meaning and final conclusion.
10 marks	Full module-style answer with diagram, derivation, design steps, assumptions and final boxed result.

Answer Writing Frame

For a 5-mark answer, write:

$$\boxed{\text{Intro} \rightarrow \text{Diagram} \rightarrow \text{Working} \rightarrow \text{Formula} \rightarrow \text{Conclusion}}$$

For an 8 or 10-mark answer, write:

$$\boxed{\text{Concept} \rightarrow \text{Neat diagram} \rightarrow \text{Each block explanation} \rightarrow \text{Derivation} \rightarrow \text{Assumptions} \rightarrow \text{Final result}}$$

3. Complete Subject Map

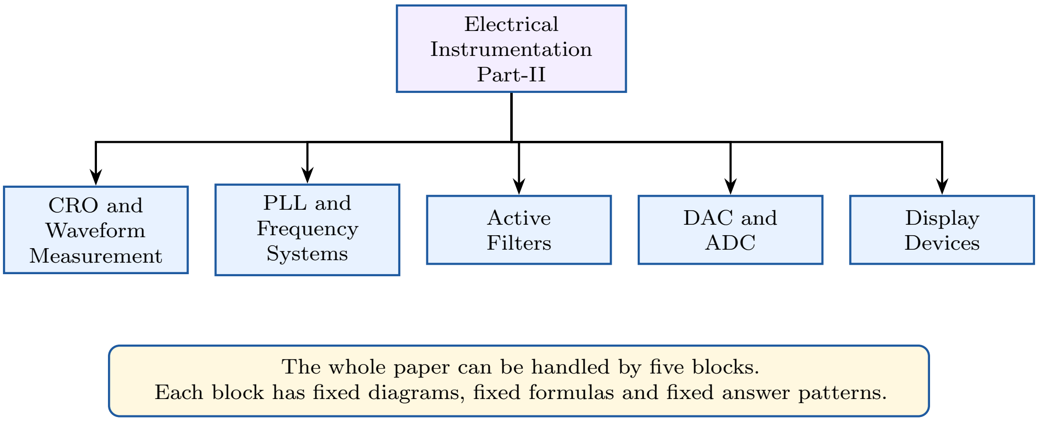

Figure 1: Reverse-engineered map of Electrical Instrumentation Part-II.

4. Question Recognition Table

Question Keyword	Topic	Immediate Response
Lissajous, tangent, frequency	CRO	Draw Lissajous idea, use tangent formula or frequency ratio.
Matched probe, compensation	CRO	Draw R-C compensation network and derive compensation condition.
Delay line	CRO	Explain why vertical signal is delayed before horizontal sweep reaches plates.
Time-base generator	CRO	Draw sweep generator and explain sawtooth voltage.
PLL stages	PLL	Draw phase detector, LPF, VCO and feedback.
Linear model of PLL	PLL	Use K_d, $F(s)$, K_v/s, derive closed-loop transfer function.
Frequency synthesizer	PLL	Draw divide-by-N feedback and use $f_o = N f_{\text{ref}}$.
Butterworth poles	Filter	Show poles lie on unit circle.
Chebyshev cut-off	Filter	Use $\|H(j1)\|^2 = 1/(1+\epsilon^2)$.
State-variable filter	Filter	Draw summer plus two integrators, write universal filter.
Switched capacitor	Filter	Derive $R_{\text{eq}} = 1/(f_s C_s)$.
R-2R ladder	DAC	Draw ladder and write binary weighted output.
Linearity error	DAC	Draw transfer characteristic and define deviation.
SAR ADC	ADC	Draw block diagram and explain MSB-first binary search.
Dual-slope ADC	ADC	Draw integrator timing diagram and derive $T_2 = (V_i/V_{\text{ref}})T_1$.
CRT, LCD, LED	Display	Draw constructional diagram and explain working block by block.

5. Universal Answer-Building Algorithm

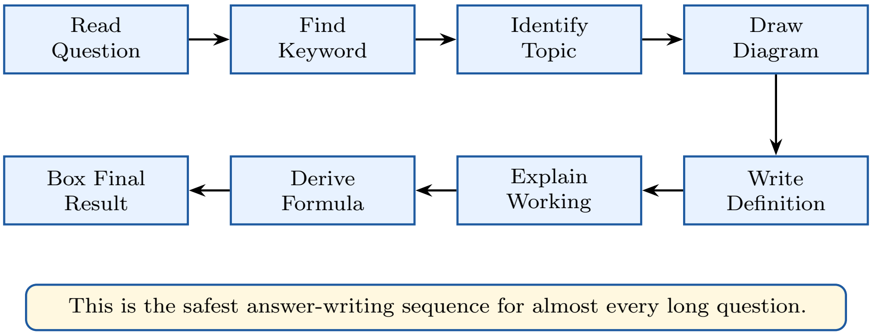

Figure 2: Universal reverse-engineering algorithm for semester answers.

Semester Command

In the exam hall, never start a long answer with random explanation. Start with definition and diagram. The diagram gives direction to the whole answer.

6. Diagram Drawing Protocol

6.1. Protocol for Block Diagrams

Use this structure:

$$\boxed{\text{Input} \rightarrow \text{Main block} \rightarrow \text{Processing block} \rightarrow \text{Output}}$$

Feedback, if present, must go below the main line.

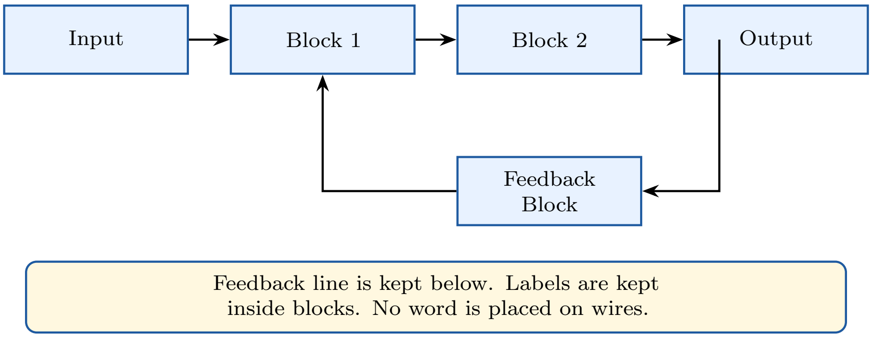

Figure 3: Safe block-diagram layout.

6.2. Protocol for Layer Diagrams

For LCD, CRT screen, or material-layer diagrams:

1. Keep actual layers at the centre.
2. Put labels outside the layer stack.
3. Use leader lines from layer to label.
4. Put light or signal arrows away from label columns.
5. If one label is at mid-height, move arrow label slightly above or below.

7. OR Question Selection Strategy

In the paper, many questions contain OR parts. You must not choose emotionally. Choose strategically.

7.1. How to Choose

Choose the option that has:

1. a diagram you can draw confidently,
2. a derivation you can complete,
3. a formula you remember,
4. fewer uncertain assumptions,
5. more standard answer structure.

If OR Contains	Choose This If
PLL linear model vs frequency synthesizer	Choose linear model if you remember $K_d F(s) K_v / s$. Choose synthesizer if numerical is simple.
Butterworth proof vs Chebyshev proof	Choose Butterworth if time is less. Choose Chebyshev only if ellipse derivation is clear.
SAR ADC vs dual-slope ADC	Choose SAR for block diagram and working. Choose dual-slope if derivation is asked.
CRT vs LCD	Choose CRT if you can draw construction. Choose LCD if question is short note or comparison.
Band-pass numerical vs filter theory	Choose numerical if values are clear and formula matches standard form.

Must Know

A 90% complete standard answer is better than a 40% complete difficult answer.

8. Core Derivations You Must Rebuild

8.1. PLL Closed-Loop Transfer Function

$$\phi_e(s) = \theta_i(s) - \theta_o(s)$$

$$V_e(s) = K_d\phi_e(s)$$

$$V_d(s) = F(s)V_e(s)$$

$$\theta_o(s) = \frac{K_v}{s}V_d(s)$$

Therefore,

$$\theta_o(s) = \frac{K_dK_vF(s)}{s}[\theta_i(s) - \theta_o(s)]$$

Let,

$$K = K_dK_v$$

Then,

$$\theta_o(s) = \frac{KF(s)}{s}[\theta_i(s) - \theta_o(s)]$$

$$s\theta_o(s) = KF(s)\theta_i(s) - KF(s)\theta_o(s)$$

$$\theta_o(s)[s + KF(s)] = KF(s)\theta_i(s)$$

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}$$

8.2. Switched-Capacitor Equivalent Resistance

Charge transferred in one clock cycle:

$$\Delta Q = C_s(V_1 - V_2)$$

Clock period:

$$T_s = \frac{1}{f_s}$$

Average current:

$$I_{\text{avg}} = \frac{\Delta Q}{T_s}$$

$$I_{\text{avg}} = C_s(V_1 - V_2)f_s$$

For equivalent resistance,

$$I = \frac{V_1 - V_2}{R_{\text{eq}}}$$

Equating,

$$\frac{V_1 - V_2}{R_{\text{eq}}} = C_s(V_1 - V_2)f_s$$

$$\boxed{R_{\text{eq}} = \frac{1}{f_s C_s}}$$

8.3. Dual-Slope ADC Derivation

First integration:

$$V_o(T_1) = -\frac{V_i T_1}{RC}$$

Second integration with reference voltage:

$$\frac{V_i T_1}{RC} = \frac{V_{\text{ref}} T_2}{RC}$$

Cancel RC:

$$V_i T_1 = V_{\text{ref}} T_2$$

$$\boxed{T_2 = \frac{V_i}{V_{\text{ref}}} T_1}$$

Since counted pulses are proportional to T_2,

$$\boxed{N = f_{\text{clk}} T_2}$$

9. Formula Bank for Last-Day Revision

9.1. CRO

$$\boxed{\frac{f_v}{f_h} = \frac{\text{number of horizontal tangencies}}{\text{number of vertical tangencies}}}$$

$$\boxed{C_m = \frac{C_i + C_p}{R_p/R_i}}$$

$$\boxed{\text{Delay line allows vertical signal to reach plates after trigger action.}}$$

9.2. PLL

$$\boxed{\phi_e = \theta_i - \theta_o}$$

$$\boxed{V_e = K_d \phi_e}$$

$$\boxed{\omega_o = \omega_{FR} + K_v V_d}$$

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}$$

$$\boxed{f_o = N f_{\text{ref}}}$$

9.3. Filters

$$\boxed{H_{BP}(s) = \frac{KBs}{s^2 + Bs + \omega_0^2}}$$

$$\boxed{H_{BS}(s) = \frac{K(s^2 + \omega_0^2)}{s^2 + Bs + \omega_0^2}}$$

$$\boxed{Q = \frac{\omega_0}{B}}$$

$$\boxed{R_{\text{eq}} = \frac{1}{f_s C_s}}$$

9.4. DAC and ADC

$$\boxed{V_{\text{LSB}} = \frac{V_{\text{ref}}}{2^n}}$$

$$\boxed{V_o = V_{\text{LSB}} D}$$

$$\boxed{\text{quantization error} = \pm\frac{1}{2}\text{LSB}}$$

$$\boxed{T_C = nT_{\text{clk}} \quad \text{for SAR ADC}}$$

$$\boxed{T_2 = \frac{V_i}{V_{\text{ref}}} T_1}$$

10. Most Important Diagrams to Practise

Topic	**Diagram Must Practise**
CRO	CRO block diagram, time-base generator, matched probe, delay line.
PLL	Basic PLL block diagram, linear model, demodulator, synthesizer.
Filters	Butterworth unit circle, Chebyshev ellipse, state-variable filter, switched-capacitor resistor.
DAC	R-2R ladder, DAC staircase, DAC error curve.
ADC	SAR ADC block diagram, SAR flowchart, dual-slope ADC timing diagram.
Display	CRT construction, electron gun, LED symbol, seven-segment display, LCD construction.

Strict Diagram Rule

Practise diagrams with a ruler-like mindset: straight line, right-angle turn, label outside, no crossing of text and wire.

11. Two-Day Execution Plan

11.1. Day 1: Concept Recovery Day

Time Block	Work
2 hours	CRO: Lissajous, matched probe, delay line and time-base generator.
2 hours	PLL: basic blocks, linear model, demodulator and synthesizer.
2.5 hours	Filters: Butterworth, Chebyshev, state-variable and switched-capacitor.
1.5 hours	Draw all diagrams once without looking.
1 hour	Revise formulas and box final results.

11.2. Day 2: Exam Simulation Day

Time Block	Work
2 hours	DAC and ADC: R-2R ladder, errors, SAR, dual-slope.
1.5 hours	Display devices: CRT, LED, LCD and comparison.
2 hours	Solve one mixed paper by selecting OR questions strategically.
1 hour	Check diagrams for overlap, missing labels and wrong arrows.
1 hour	Final formula bank and derivation bank revision.

Semester Command

The last night is not for learning new theory. It is for diagram memory, formula memory and answer structure memory.

12. Answer Templates

12.1. Template A: Diagram-Based Theory Question

Answer Writing Frame

Step 1: Definition.
The given system is defined as ...
Step 2: Diagram.
Draw the labelled diagram neatly.
Step 3: Working.
Explain the function of each block from input to output.
Step 4: Important Formula.
Write the main formula and explain the symbols.
Step 5: Conclusion.
Therefore, the system performs ...

12.2. Template B: Derivation Question

Answer Writing Frame

Step 1: State assumptions.
Assume ideal components, linear operation and standard notation.
Step 2: Write starting equation.
Begin from the physical relation.
Step 3: Substitute block relations.
Use the relation of each block.
Step 4: Rearrange carefully.
Bring output terms to one side.
Step 5: Box final result.

$$\boxed{\text{Final expression}}$$

12.3. Template C: Numerical Question

Answer Writing Frame

Step 1: Given data.
Write all given quantities.
Step 2: Required quantity.
Write what is to be found.
Step 3: Formula.
Select the direct formula.
Step 4: Substitution.
Substitute values with units.
Step 5: Final answer.
Box the answer.

13. Final Self-Test Questions

13.1. CRO

1. Can I draw the CRO vertical and horizontal system?
2. Can I explain why delay line is used?
3. Can I solve Lissajous tangent ratio problems?
4. Can I derive matched probe compensation condition?

13.2. PLL

1. Can I draw PLL block diagram without looking?
2. Can I derive $\theta_o(s)/\theta_i(s)$?
3. Can I explain capture range and lock range?
4. Can I solve $f_o = N f_{\text{ref}}$?

13.3. Filters

1. Can I prove Butterworth poles lie on a unit circle?
2. Can I prove Chebyshev poles lie on an ellipse?
3. Can I write band-pass and band-stop transfer functions?
4. Can I derive $R_{\text{eq}} = 1/(f_s C_s)$?

13.4. Converters

1. Can I draw R-2R ladder?
2. Can I explain DAC linearity error?
3. Can I draw SAR ADC and explain binary search?
4. Can I derive dual-slope ADC relation?

13.5. Displays

1. Can I draw CRT construction?
2. Can I explain electron gun?
3. Can I draw seven-segment display?
4. Can I draw LCD without label overlap?

14. Final Exam-Hall Command Sheet

Semester Command

Before answering each question, silently ask:

Is this a diagram question, derivation question, comparison question or numerical question?

Then choose the suitable template.

Must Know

Your best scoring style:

neat heading + definition + clean diagram + block explanation + boxed formula

Strict Diagram Rule

Never do these in the exam:

1. Do not draw diagrams without labels.
2. Do not write labels over arrows.
3. Do not skip final formula in derivation.
4. Do not leave OR decision until the middle of the answer.
5. Do not spend too much time beautifying one diagram.

Phase 6 Completed: Final Reverse-Engineering Mastery